Population Attributable Fraction of Obesity for Hypercapnic Respiratory Failure

Alyssa Schreck MD, Brian Locke MD MSc

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1 Obesity and hypercapnic respiratory failure analysis

1.1 Section 0. Setup

1.1.1 Load packages

```
library(haven)
library(ggplot2)
library(modelsummary)
```

```
Warning: package 'modelsummary' was built under R version 4.5.3
```

```
library(here)
```

```
Warning: package 'here' was built under R version 4.5.3
```

```
here() starts at C:/Users/ajsch/Downloads
```

```
library(gtsummary)
library(stringr)
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

1.1.2 Define file paths and constants

```r
data_path <- here::here("full_db.rds")

co2_abg_cutoff <- 45
co2_vbg_cutoff <- 50
ph_cutoff <- 7.35
icd_hrf_vars <- c("has_j9612", "has_j9622", "has_j9692", "has_j9602")

coef_map_common <- c(
  "curr_bmi_num" = "BMI (per 1 kg/m^2)",
  "age_at_encounter" = "Age (per year)",
  "sexMale" = "Male",
  "ethnicityHispanic or Latino" = "Hispanic/Latino",
  "ethnicityUnknown" = "Ethnicity: Unknown"
)
```

1.2 Section 1. Data ingest and harmonization

1.2.1 readRDS()

```r
stata_data <- readRDS(data_path)
```

1.2.2 Standardize types (labelled -> numeric, factors)

```r
# Verify before conversion
table(stata_data$niv_proc, useNA = "ifany")
```

```
                                  No NIV
                                  799385
Received NIV (not including nightly CPAP)
                                   34091
```

```r
# Convert labelled variables to numeric properly
stata_data <- stata_data %>%
  mutate(
    niv_proc = case_when(
      as.character(as_factor(niv_proc)) == "Received NIV (not including nightly CPAP)" ~ 1,
      as.character(as_factor(niv_proc)) == "No NIV" ~ 0,
      TRUE ~ NA_real_
    ),
    imv_proc = case_when(
      as.character(as_factor(imv_proc)) == "Received IMV" ~ 1,
      as.character(as_factor(imv_proc)) == "No IMV" ~ 0,
      TRUE ~ NA_real_
    ),
    highest_paco2 = as.numeric(highest_paco2),
    abg_ph = as.numeric(abg_ph),
    vbg_co2 = as.numeric(vbg_co2),
    vbg_ph = as.numeric(vbg_ph)
  )
```

```r
# Verify after conversion
table(stata_data$niv_proc, useNA = "ifany")
```

```
     0      1
799385  34091
```

```r
stopifnot(all(is.na(stata_data$niv_proc) | stata_data$niv_proc %in% c(0, 1)))
stopifnot(any(stata_data$niv_proc == 0, na.rm = TRUE))
stopifnot(any(stata_data$niv_proc == 1, na.rm = TRUE))

stopifnot(all(is.na(stata_data$imv_proc) | stata_data$imv_proc %in% c(0, 1)))
stopifnot(any(stata_data$imv_proc == 0, na.rm = TRUE))
stopifnot(any(stata_data$imv_proc == 1, na.rm = TRUE))

any_vent <- (stata_data$niv_proc == 1 | stata_data$imv_proc == 1)
stopifnot(any(any_vent == TRUE, na.rm = TRUE))
stopifnot(any(any_vent == FALSE, na.rm = TRUE))
```

1.2.3 Quick sanity checks (row counts, missingness)

```r
nrow(stata_data)
```

```
[1] 833476
```

TODO: Add missingness checks.

```r
library(dplyr)

stata_data %>%
  summarise(
    total_encounters = n(),
```

```
    niv_count = sum(niv_proc == 1, na.rm = TRUE),
    imv_count = sum(imv_proc == 1, na.rm = TRUE),
    any_vent = sum(niv_proc == 1 | imv_proc == 1, na.rm = TRUE),
    neither = sum(niv_proc == 0 & imv_proc == 0, na.rm = TRUE)
  )
```

total_encounters	niv_count	imv_count	any_vent	neither
833476	34091	58232	85678	747798

BMI missingness:

```
table(ifelse(is.na(stata_data$curr_bmi), "BMI Missing (NA)", "BMI Value Present"))
```

```
 BMI Missing (NA) BMI Value Present
           450715             382761
```

VBG, ABG, and CO2 missingness:

```
library(dplyr)
library(tidyr)

missing_summary <- tibble(
  variable = c("VBG CO2", "Highest PaCO2", "ABG pH", "VBG pH"),
  missing = c(
    sum(is.na(stata_data$vbg_co2)),
    sum(is.na(stata_data$highest_paco2)),
    sum(is.na(stata_data$abg_ph)),
    sum(is.na(stata_data$vbg_ph))
  ),
  present = c(
    sum(!is.na(stata_data$vbg_co2)),
    sum(!is.na(stata_data$highest_paco2)),
```

```
    sum(!is.na(stata_data$abg_ph)),
    sum(!is.na(stata_data$vbg_ph))
  )
) %>%
  mutate(
    total = missing + present,
    pct_missing = round(100 * missing / total, 1)
  )

missing_summary
```

variable	missing	present	total	pct_missing
VBG CO2	677009	156467	833476	81.2
Highest PaCO2	635606	197870	833476	76.3
ABG pH	633941	199535	833476	76.1
VBG pH	666142	167334	833476	79.9

```
table(
  "VBG CO2 & VBG pH availability" =
    ifelse(
      is.na(stata_data$vbg_co2) | is.na(stata_data$vbg_ph),
      "One or Both Missing",
      "Both Present"
    )
)
```

```
VBG CO2 & VBG pH availability
       Both Present One or Both Missing
             156399              677077
```

```
table(
  "Highest PaCO2 & ABG pH availability" =
    ifelse(
```

```
      is.na(stata_data$highest_paco2) | is.na(stata_data$abg_ph),
      "One or Both Missing",
      "Both Present"
    )
)
```

```
Highest PaCO2 & ABG pH availability
       Both Present One or Both Missing
             197141              636335
```

```
stata_data %>%
  summarise(
    bmi_min = min(curr_bmi, na.rm = TRUE),
    bmi_max = max(curr_bmi, na.rm = TRUE),
    paco2_min = min(highest_paco2, na.rm = TRUE),
    paco2_max = max(highest_paco2, na.rm = TRUE),
    ph_min = min(abg_ph, na.rm = TRUE),
    ph_max = max(abg_ph, na.rm = TRUE)
  )
```

bmi_min	bmi_max	paco2_min	paco2_max	ph_min	ph_max
10.09	50	6	249	5.912479	8.011059

1.3 Section 2. Variable derivations

1.3.1 BMI parsing + BMI categories

```
bmi_raw <- stata_data %>%
  transmute(curr_bmi_raw = na_if(str_trim(as.character(zap_labels(curr_bmi))), "n/a"))

non_numeric_bmi <- bmi_raw %>%
  filter(!is.na(curr_bmi_raw), !str_detect(curr_bmi_raw, "^\\d+(\\.\\d+)?$"))
```

```r
non_numeric_bmi %>%
  count(curr_bmi_raw, sort = TRUE)
```

curr_bmi_raw	n

```r
non_numeric_bmi %>%
  summarise(n_failed_parse = n())
```

n_failed_parse
0

```r
stata_data <- stata_data %>%
  mutate(
    curr_bmi_num = if_else(
      str_detect(str_trim(as.character(zap_labels(curr_bmi))), "^\\d+(\\.\\d+)?$"),
      as.numeric(str_trim(as.character(zap_labels(curr_bmi)))),
      NA_real_
    )
  )
```

```r
stata_data <- stata_data %>%
  mutate(
    bmi_category = case_when(
      is.na(curr_bmi_num) ~ NA_character_,
      curr_bmi_num < 18.5 ~ "Underweight",
      curr_bmi_num < 25   ~ "Normal weight",
      curr_bmi_num < 30   ~ "Overweight",
      curr_bmi_num < 35   ~ "Obesity I",
      curr_bmi_num < 40   ~ "Obesity II",
      curr_bmi_num >= 40  ~ "Obesity III"
    )
  )
```

1.3.2 HRF definition v1 (gases + ventilation)

Below is my first (probably most stringent) iteration of hypercapnic respiratory failure: co2 greater than 45 if abg, 50 if vbg, ph <7.35 - they have to have either an abg or a vbg showing this, AND either NIV or IMV or both. I've written it so that ABG values and VBG values cannot mix

```
stata_data <- stata_data %>%
  mutate(
    gas_source = case_when(
      !is.na(highest_paco2) & !is.na(abg_ph) ~ "ABG",
      TRUE ~ "VBG"
    )
  )
```

```
stata_data <- stata_data %>%
  mutate(
    hrf_hypercapnic = case_when(
      # ABG-based HRF
      (niv_proc == 1 | imv_proc == 1) &
        gas_source == "ABG" &
        !is.na(highest_paco2) & !is.na(abg_ph) &
        highest_paco2 >= co2_abg_cutoff &
        abg_ph <= ph_cutoff ~ 1,

      # VBG-based HRF
      (niv_proc == 1 | imv_proc == 1) &
        gas_source == "VBG" &
        !is.na(vbg_co2) & !is.na(vbg_ph) &
        vbg_co2 >= co2_vbg_cutoff &
        vbg_ph <= ph_cutoff ~ 1,

      # Explicit non-HRF ONLY if gases exist
      (gas_source == "ABG" & !is.na(highest_paco2) & !is.na(abg_ph)) |
      (gas_source == "VBG" & !is.na(vbg_co2) & !is.na(vbg_ph)) ~ 0,

      TRUE ~ NA_real_
```

```
    ),
    hrf_hypercapnic = factor(
      hrf_hypercapnic,
      levels = c(0, 1),
      labels = c("No", "Yes")
    )
  )
```

*changed the above from TRUE ~ 0 to TRUE ~ NA_real

1.3.3 HRF definition v2 (labs + ICD) and v2_strict

Regression done with new definition of hypercapnic respiratory failure - pH and CO2 definition as above, except now including (required) ICD codes instead of ventilation (I only used 4 - the first four that are listed in database):

```
stata_data <- stata_data %>%
  mutate(
    # Lab-based hypercapnic respiratory failure (ABG or VBG)
    hrf_hypercapnic_labs_v2 = case_when(
      !is.na(highest_paco2) & !is.na(abg_ph) &
       highest_paco2 >= co2_abg_cutoff & abg_ph < ph_cutoff ~ 1, # BL note - fixed this 50->45

      !is.na(vbg_co2) & !is.na(vbg_ph) &
        vbg_co2 >= co2_vbg_cutoff & vbg_ph < ph_cutoff ~ 1,

      TRUE ~ 0
    ),

    # ICD-based hypercapnic respiratory failure
    hrf_hypercapnic_icd_v2 = ifelse(
      rowSums(across(all_of(icd_hrf_vars)) == 1, na.rm = TRUE) > 0,
      1, 0
    ),
```

```
    # Final combined HRF definition (labs OR ICD)
    hrf_hypercapnic_v2 = ifelse(
      hrf_hypercapnic_labs_v2 == 1 |
      hrf_hypercapnic_icd_v2 == 1,
      1, 0
    )
  )
```

```
stata_data <- stata_data %>%
  mutate(
    hrf_hypercapnic_v2_strict = ifelse(
      hrf_hypercapnic_labs_v2 == 1 &
      hrf_hypercapnic_icd_v2  == 1,
      1, 0
    )
  )
```

TODO: consider also restricting non HRF HERE (rather than in primary analysis cohort)

*Non-HRF group definition (has BMI and gas recorded, just without meeting criteria for HRF):

```
stata_data <- stata_data %>%
  mutate(
    has_complete_gas = case_when(
      gas_source == "ABG" ~ !is.na(highest_paco2) & !is.na(abg_ph),
      gas_source == "VBG" ~ !is.na(vbg_co2) & !is.na(vbg_ph),
      TRUE ~ FALSE
    ),

    non_hrf = case_when(
      has_complete_gas == TRUE & hrf_hypercapnic == "No" ~ 1,
      has_complete_gas == TRUE & hrf_hypercapnic == "Yes" ~ 0,
      TRUE ~ NA_real_
    ),
```

```
    non_hrf = factor(
      non_hrf,
      levels = c(0, 1),
      labels = c("HRF", "Non-HRF")
    )
  )
```

*Separate definition for hypercapnia without acidosis or respiratory failure (note i have not yet pulled this group out of the non HRF group, so theoretically these patients are a subgroup of the above group):

```
stata_data <- stata_data %>%
  mutate(
    compensated_hypercapnia = case_when(
      gas_source == "ABG" &
        !is.na(highest_paco2) & !is.na(abg_ph) &
        highest_paco2 >= co2_abg_cutoff &
        abg_ph > ph_cutoff ~ 1,

      gas_source == "VBG" &
        !is.na(vbg_co2) & !is.na(vbg_ph) &
        vbg_co2 >= co2_vbg_cutoff &
        vbg_ph > ph_cutoff ~ 1,

      TRUE ~ 0
    )
  )
```

1.4 Section 3. Cohort construction and attrition

1.4.1 Primary analysis cohort (HRF v1)

```
stata_hrf_analysis <- stata_data %>%
  filter(!is.na(curr_bmi_num),
```

```
        (!is.na(highest_paco2) & !is.na(abg_ph)) |
        (!is.na(vbg_co2) & !is.na(vbg_ph)))
```

1.4.2 Strict v2 cohort (sensitivity)

```
stata_hrf_strict_analysis <- stata_data %>%
  filter(
    !is.na(curr_bmi_num),
    has_complete_gas == TRUE,
    (
      hrf_hypercapnic_v2_strict == 1 |
      (
        hrf_hypercapnic_labs_v2 == 0 &
        hrf_hypercapnic_icd_v2  == 0
      )
    )
  )
```

* Secondary cohort

- Here I created a cohort for a possible secondary analysis - I stratified first by hypercapnia, and then defined two outcomes - HRF vs compensated hypercapnia for use later for a third regression

```
stata_data <- stata_data %>% mutate(
  hypercapnia_flag = case_when(
    (!is.na(highest_paco2) & highest_paco2 >= co2_abg_cutoff) ~ 1,
    (!is.na(vbg_co2) & vbg_co2 >= co2_vbg_cutoff) ~ 1,
    TRUE ~ 0
  )
)
```

```
hypercapnia_cohort <- stata_data %>%
  filter(
    has_complete_gas == TRUE,
```

```
    hypercapnia_flag == TRUE
  )
```

```
hypercapnia_cohort <- hypercapnia_cohort %>%
  mutate(
    hrf_vs_comp = case_when(
      hrf_hypercapnic == "Yes" ~ 1,
      compensated_hypercapnia == 1 ~ 0,
      TRUE ~ NA_real_
    )
  )
```

TODO: Add attrition table with inclusion/exclusion counts.

```
attrition_primary <- tibble(
  step = c(
    "Initial dataset",
    "Exclude missing BMI",
    "Exclude missing ABG/VBG gases",
    "Final primary cohort"
  ),
  n = c(
    nrow(stata_data),
    sum(stata_data$has_bmi),
    sum(stata_data$has_bmi & stata_data$has_any_gas),
    nrow(stata_hrf_analysis)
  )
)
```

```
Warning: Unknown or uninitialised column: `has_bmi`.
Unknown or uninitialised column: `has_bmi`.
```

```
Warning: Unknown or uninitialised column: `has_any_gas`.
```

```
attrition_primary
```

step	n
Initial dataset	833476
Exclude missing BMI	0
Exclude missing ABG/VBG gases	0
Final primary cohort	112009

1.5 Section 4. Descriptive statistics

1.5.1 Distribution of BMI in the analytic cohort

```
table(stata_data$hrf_hypercapnic, useNA = "ifany")
```

```
    No    Yes   <NA>
271633  25774 536069
```

```
summary(stata_data$curr_bmi_num)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
  10.09   26.31   33.34   33.64   41.20   50.00  450715
```

```
table(stata_data$bmi_category, useNA = "ifany")
```

```
Normal weight     Obesity I    Obesity II   Obesity III    Overweight
        65912         58317         44022        130477         74499
  Underweight          <NA>
         9534        450715
```

[BL - note high rates of BMI missing data]

1.5.2 HRF prevalence by BMI category (v1)

Counts (bottom right is amount of patients meeting criteria above- note* now no longer adds up to full dataset, instead it should add up to the number of patients with classifiable HRF status

```
table(
  hrf = stata_data$hrf_hypercapnic,
  any_vent = (stata_data$niv_proc == 1 | stata_data$imv_proc == 1)
)
```

```
     any_vent
hrf     FALSE   TRUE
  No   229786  41847
  Yes       0  25774
```

1.5.3 Counts for both HRF as defined above AND current BMI above 30

```
stata_data %>%
  filter(
    hrf_hypercapnic == "Yes",
    curr_bmi_num >= 30
  ) %>%
  summarise(
    n_patients = n()
  )
```

n_patients
4099

Counts for HRF and BMI by category

```r
stata_data %>%
  filter(
    hrf_hypercapnic == "Yes",
    !is.na(curr_bmi_num)
  ) %>%
  mutate(
    bmi_category = case_when(
      curr_bmi_num < 18.5 ~ "Underweight",
      curr_bmi_num < 25   ~ "Normal weight",
      curr_bmi_num < 30   ~ "Overweight",
      curr_bmi_num < 35   ~ "Obesity I",
      curr_bmi_num < 40   ~ "Obesity II",
      curr_bmi_num >= 40  ~ "Obesity III"
    )
  ) %>%
  count(bmi_category) %>%
  mutate(pct = round(100 * n / sum(n), 1))
```

bmi_category	n	pct
Normal weight	2311	25.1
Obesity I	1766	19.2
Obesity II	1131	12.3
Obesity III	1202	13.1
Overweight	2287	24.9
Underweight	503	5.5

1.5.4 Baseline characteristics table

☐ TODO

*(need to re-work - needs some work with the categories, cleaning up variables)

```r
library(dplyr)
library(gtsummary)

tbl1 <- stata_data %>%
  mutate(
    # Outcome grouping
    hrf_group = case_when(
      hrf_hypercapnic == "Yes" ~ "HRF",
      hrf_hypercapnic == "No"  ~ "No HRF",
      TRUE ~ NA_character_
    ),

    # BMI grouping
    bmi_group = case_when(
      curr_bmi_num < 25 ~ "Normal",
      curr_bmi_num < 30 ~ "Overweight",
      curr_bmi_num >= 30 ~ "Obese",
      TRUE ~ "Missing BMI"
    )
  ) %>%
  select(
    hrf_group,
    bmi_group,
    curr_bmi_num,
    age_at_encounter,
    sex,
    ethnicity,
    highest_paco2,
    abg_ph,
    vbg_co2,
    vbg_ph,
    niv_proc,
    imv_proc
  ) %>%
  tbl_summary(
    by = hrf_group,
```

```
    missing = "ifany",
    statistic = list(
      all_continuous() ~ "{mean} ({sd})",
      all_categorical() ~ "{n} ({p}%)"
    )
  ) %>%
  add_overall()
```

```
536069 missing rows in the "hrf_group" column have been removed.
```

```
tbl1
```

1.5.5 Are proportions of hypercapnic respiratory failure (v1) different across BMI categories?

Just out of curiosity ran Chi square analysis below

```
bmi_hrf_subset <- stata_hrf_analysis %>%
  filter(!is.na(bmi_category))
```

```
bmi_hrf_summary <- bmi_hrf_subset %>%
  group_by(bmi_category) %>%
  summarise(
    n_total = n(),
    n_hrf = sum(hrf_hypercapnic == "Yes"),
    n_no_hrf = sum(hrf_hypercapnic == "No"),
    pct_hrf = round(100 * n_hrf / n_total, 1),
    pct_no_hrf = round(100 * n_no_hrf / n_total, 1)
  ) %>%
  arrange(factor(bmi_category, levels = c("Underweight", "Normal weight", "Overweight",
                                          "Obesity I", "Obesity II", "Obesity III")))

chisq_test <- chisq.test(table(bmi_hrf_subset$bmi_category, bmi_hrf_subset$hrf_hypercapnic))

bmi_hrf_summary
```

Characteristic	**Overall** N = 297,407[1]	**HRF** N = 25,774[1]	**No HRF** N = 271,633[1]
bmi_group			
Missing BMI	185,398 (62%)	16,574 (64%)	168,824 (62%)
Normal	36,121 (12%)	2,814 (11%)	33,307 (12%)
Obese	44,132 (15%)	4,099 (16%)	40,033 (15%)
Overweight	31,756 (11%)	2,287 (8.9%)	29,469 (11%)
curr_bmi_num	29 (7)	30 (8)	29 (7)
Unknown	185,398	16,574	168,824
age_at_encounter	60 (17)	61 (16)	60 (17)
Sex			
Female	137,949 (46%)	10,763 (42%)	127,186 (47%)
Male	159,458 (54%)	15,011 (58%)	144,447 (53%)
Unknown or Other	0 (0%)	0 (0%)	0 (0%)
Ethnicity			
Not Hispanic or Latino	212,679 (72%)	20,459 (79%)	192,220 (71%)
Hispanic or Latino	19,229 (6.5%)	1,681 (6.5%)	17,548 (6.5%)
Unknown	65,499 (22%)	3,634 (14%)	61,865 (23%)
highest_paco2	49 (22)	70 (31)	46 (19)
Unknown	99,737	1,932	97,805
abg_ph	7.37 (0.11)	7.23 (0.11)	7.38 (0.10)
Unknown	98,956	1,826	97,130
vbg_co2	46 (13)	56 (19)	45 (11)
Unknown	140,999	13,857	127,142
vbg_ph	7.35 (0.11)	7.25 (0.14)	7.35 (0.10)
Unknown	136,646	12,988	123,658
niv_proc	22,684 (7.6%)	9,359 (36%)	13,325 (4.9%)
imv_proc	50,846 (17%)	19,105 (74%)	31,741 (12%)

[1] n (%); Mean (SD)

bmi_category	n_total	n_hrf	n_no_hrf	pct_hrf	pct_no_hrf
Underweight	5061	503	4558	9.9	90.1
Normal weight	31060	2311	28749	7.4	92.6
Overweight	31756	2287	29469	7.2	92.8
Obesity I	21692	1766	19926	8.1	91.9
Obesity II	12223	1131	11092	9.3	90.7
Obesity III	10217	1202	9015	11.8	88.2

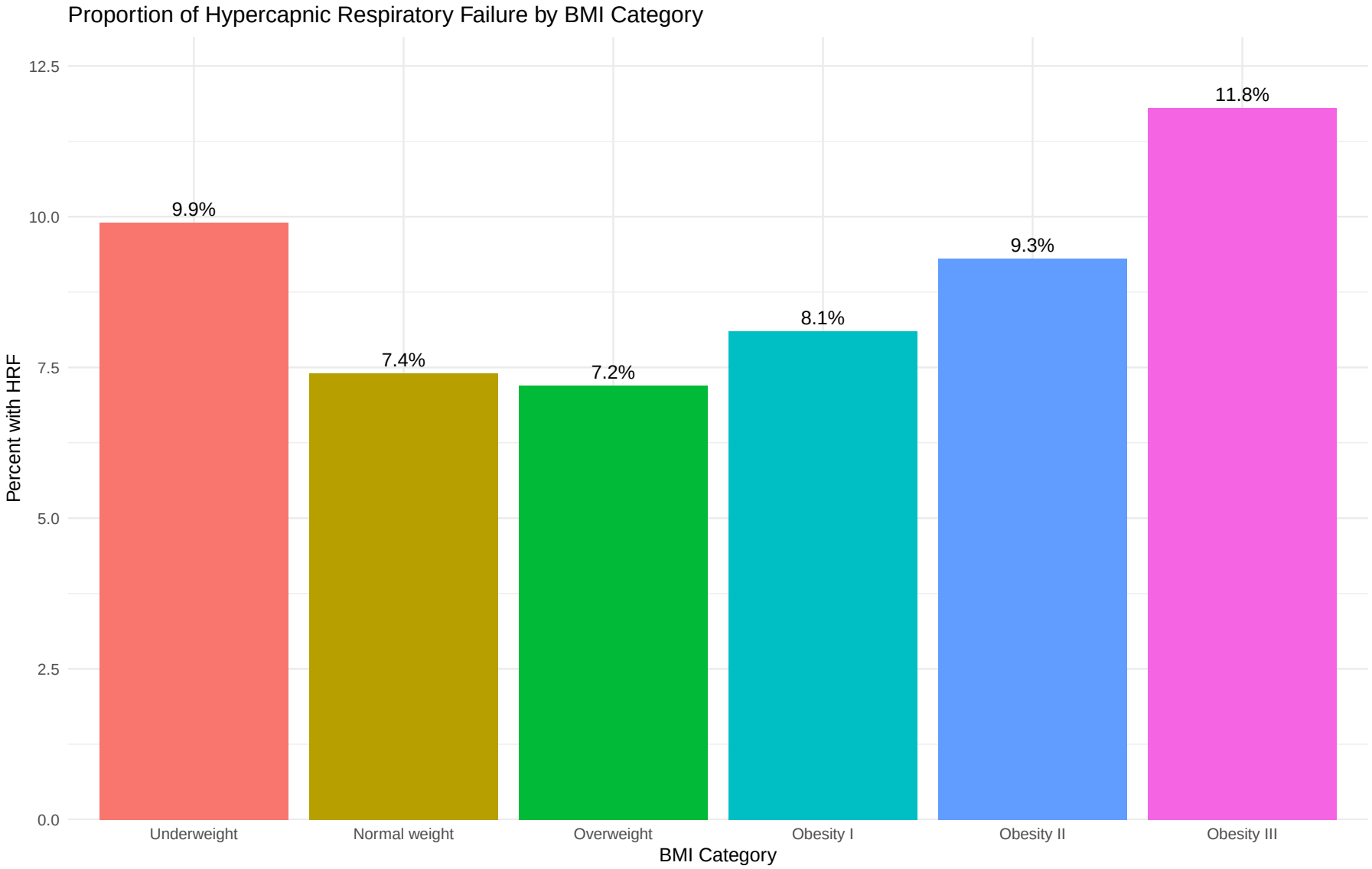
```
chisq_test$p.value
```

```
[1] 1.243203e-57
```

Visualization:

```
# Make sure bmi_category is ordered
bmi_hrf_summary <- bmi_hrf_summary %>%
  mutate(bmi_category = factor(bmi_category,
                               levels = c("Underweight", "Normal weight", "Overweight",
                                          "Obesity I", "Obesity II", "Obesity III")))

# Plot proportion of HRF by BMI category
ggplot(bmi_hrf_summary, aes(x = bmi_category, y = pct_hrf, fill = bmi_category)) +
  geom_col() +
  geom_text(aes(label = paste0(pct_hrf, "%")), vjust = -0.5) +
  scale_y_continuous(expand = expansion(mult = c(0, 0.1))) +
  labs(
    title = "Proportion of Hypercapnic Respiratory Failure by BMI Category",
    x = "BMI Category",
    y = "Percent with HRF"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```


Proportion of Hypercapnic Respiratory Failure by BMI Category
Percent with HRF
12.5
10.0
7.5
5.0
2.5
0.0
9.9%
7.4%
7.2%
8.1%
9.3%
11.8%
Underweight
Normal weight
Overweight
Obesity I
Obesity II
Obesity III
BMI Category

1.6 Section 5. Primary inferential model

1.6.1 Unadjusted (aka simple) regression for outcome 1

Below is regression, starting with BMI (the degrees of freedom is 112k which seems appropriate - includes all patients with a BMI and labs/vent indicating NO HRF, plus the patients with BMI and hypercapnic respiratory failure as defined above)

```
logit_bmi_hrf <- glm(hrf_hypercapnic ~ curr_bmi_num,
data = stata_hrf_analysis,
family = binomial)

summary(logit_bmi_hrf)
```

```
Call:
glm(formula = hrf_hypercapnic ~ curr_bmi_num, family = binomial,
    data = stata_hrf_analysis)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  -2.92776    0.04482  -65.33   <2e-16 ***
curr_bmi_num  0.01750    0.00146   11.98   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 63611  on 112008  degrees of freedom
Residual deviance: 63470  on 112007  degrees of freedom
AIC: 63474

Number of Fisher Scoring iterations: 5
```

OR shows small increase in risk with BMI, though is statistically significant

[BL Edit - though keep in mind, Odds increased by .02 for each BMI point (e.g. 10 = 20% increase in odds)]

1.6.2 Adjusted, multivariable regression for outcome 1

Regression done again adjusted for variables NOT influenced by BMI (age, sex, ethnicity included - didn't include location as it appears to be the same for all entries)

```
baseline_model <- glm(
hrf_hypercapnic ~ curr_bmi_num + age_at_encounter + sex + ethnicity,
data = stata_hrf_analysis,
family = binomial
)

summary(baseline_model)
```

```
Call:
glm(formula = hrf_hypercapnic ~ curr_bmi_num + age_at_encounter +
    sex + ethnicity, family = binomial, data = stata_hrf_analysis)

Coefficients:
                             Estimate Std. Error z value Pr(>|z|)
(Intercept)                -3.3143678  0.0653822 -50.692  < 2e-16 ***
curr_bmi_num                0.0194994  0.0014798  13.177  < 2e-16 ***
age_at_encounter            0.0028333  0.0006493   4.363 1.28e-05 ***
sexMale                     0.2388572  0.0222256  10.747  < 2e-16 ***
ethnicityHispanic or Latino -0.0444911  0.0432380  -1.029    0.303
ethnicityUnknown            0.1622532  0.0302736   5.360 8.34e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 63611  on 112008  degrees of freedom
Residual deviance: 63302  on 112003  degrees of freedom
AIC: 63314

Number of Fisher Scoring iterations: 5
```

```r
mods_v1 <- list(
  "HRF v1: unadjusted" = logit_bmi_hrf,
  "HRF v1: adjusted"   = baseline_model
)

modelsummary(
  mods_v1,
  exponentiate = TRUE,
  statistic    = "conf.int",
  coef_map = coef_map_common,
  gof_omit = "RMSE|F|Adj|R2"
)
```

```
Profiled confidence intervals may take longer time to compute.
  Use `ci_method="wald"` for faster computation of CIs.
Profiled confidence intervals may take longer time to compute.
  Use `ci_method="wald"` for faster computation of CIs.
```

Association appears to be stronger once adjusted

1.7 Section 6. Sensitivity analyses

1.7.1 Counts for strict HRF definition (v2_strict)

Number of patients meeting strict definition

```r
table(stata_data$hrf_hypercapnic_v2_strict, useNA = "ifany")
```

```
     0      1
817895  15581
```

	HRF v1: unadjusted	HRF v1: adjusted
BMI (per 1 kg/m^2)	1.018	1.020
	[1.015, 1.021]	[1.017, 1.023]
Age (per year)		1.003
		[1.002, 1.004]
Male		1.270
		[1.216, 1.326]
Hispanic/Latino		0.956
		[0.878, 1.040]
Ethnicity: Unknown		1.176
		[1.108, 1.248]
Num.Obs.	112 009	112 009
AIC	63 473.9	63 313.9
BIC	63 493.2	63 371.7
Log.Lik.	−31 734.972	−31 650.965

1.7.2 Unadjusted Logistic Regression for Outcome 2

```r
model_strict_bmi_unadj <- glm(
  hrf_hypercapnic_v2_strict ~ curr_bmi_num,
  family = binomial,
  data = stata_hrf_strict_analysis
)

summary(model_strict_bmi_unadj)
```

```
Call:
glm(formula = hrf_hypercapnic_v2_strict ~ curr_bmi_num, family = binomial,
    data = stata_hrf_strict_analysis)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  -3.449156   0.059003  -58.46   <2e-16 ***
curr_bmi_num  0.023881   0.001909   12.51   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 39644  on 87024  degrees of freedom
Residual deviance: 39491  on 87023  degrees of freedom
AIC: 39495

Number of Fisher Scoring iterations: 5
```

1.7.3 Adjusted, Multivariable Regression for Outcome 2

Then adjusted for age, sex, ethnicity:

```r
baseline_model_strict <- glm(
  hrf_hypercapnic_v2_strict ~
    curr_bmi_num +
    age_at_encounter +
    sex +
    ethnicity,
  data = stata_hrf_strict_analysis,
  family = binomial
)

summary(baseline_model_strict)
```

```
Call:
glm(formula = hrf_hypercapnic_v2_strict ~ curr_bmi_num + age_at_encounter +
    sex + ethnicity, family = binomial, data = stata_hrf_strict_analysis)

Coefficients:
                             Estimate Std. Error z value Pr(>|z|)
(Intercept)                -4.7689449  0.0935568 -50.974  < 2e-16 ***
curr_bmi_num                0.0277918  0.0019502  14.251  < 2e-16 ***
age_at_encounter            0.0194879  0.0009325  20.900  < 2e-16 ***
sexMale                    -0.0351480  0.0287347  -1.223  0.22126
ethnicityHispanic or Latino -0.1912798  0.0616342  -3.103  0.00191 **
ethnicityUnknown            0.0537704  0.0416200   1.292  0.19638
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 39644  on 87024  degrees of freedom
Residual deviance: 38980  on 87019  degrees of freedom
AIC: 38992

Number of Fisher Scoring iterations: 6
```

```r
mods_v2 <- list(
  "HRF v2 strict: unadjusted" = model_strict_bmi_unadj,
  "HRF v2 strict: adjusted"   = baseline_model_strict
)

modelsummary(
  mods_v2,
  exponentiate = TRUE,
  statistic    = "conf.int",
  coef_map = coef_map_common,
  gof_omit = "RMSE|F|Adj|R2"
)
```

```
Profiled confidence intervals may take longer time to compute.
  Use `ci_method="wald"` for faster computation of CIs.
Profiled confidence intervals may take longer time to compute.
  Use `ci_method="wald"` for faster computation of CIs.
```

```r
mods_all <- list(
  "HRF v1 (primary)"            = baseline_model,
  "HRF v2_strict (sensitivity)" = baseline_model_strict
)

modelsummary(
  mods_all,
  exponentiate = TRUE,
  statistic    = "conf.int",
  coef_map = coef_map_common,
  gof_omit = "RMSE|F|Adj|R2"
)
```

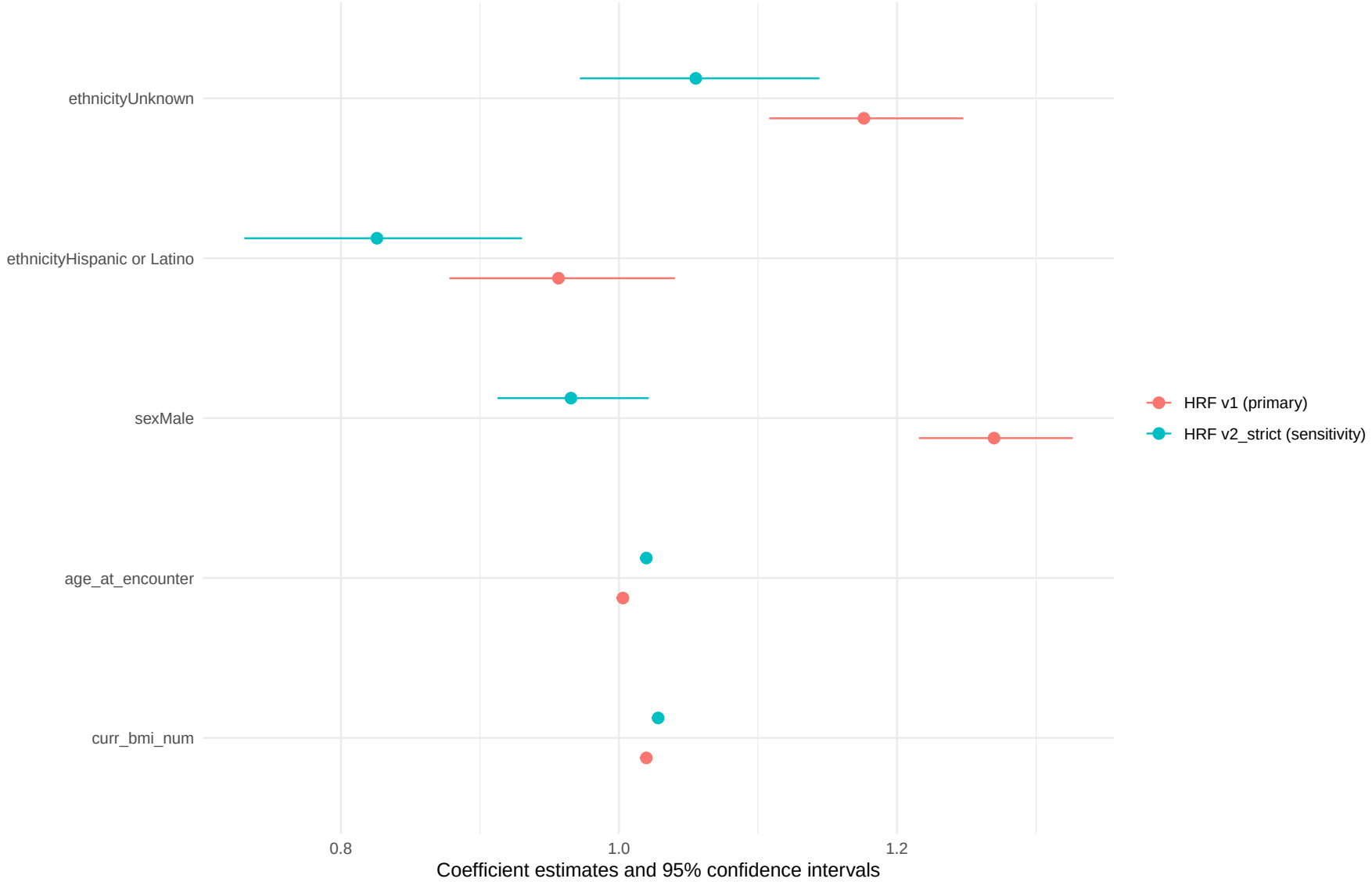
```
Profiled confidence intervals may take longer time to compute.
  Use `ci_method="wald"` for faster computation of CIs.
Profiled confidence intervals may take longer time to compute.
  Use `ci_method="wald"` for faster computation of CIs.
```

	HRF v2 strict: unadjusted	HRF v2 strict: adjusted
BMI (per 1 kg/m^2)	1.024	1.028
	[1.020, 1.028]	[1.024, 1.032]
Age (per year)		1.020
		[1.018, 1.022]
Male		0.965
		[0.913, 1.021]
Hispanic/Latino		0.826
		[0.731, 0.930]
Ethnicity: Unknown		1.055
		[0.972, 1.144]
Num.Obs.	87 025	87 025
AIC	39 495.0	38 992.1
BIC	39 513.7	39 048.3
Log.Lik.	−19 745.497	−19 490.036

	HRF v1 (primary)	HRF v2_strict (sensitivity)
BMI (per 1 kg/m^2)	1.020	1.028
	[1.017, 1.023]	[1.024, 1.032]
Age (per year)	1.003	1.020
	[1.002, 1.004]	[1.018, 1.022]
Male	1.270	0.965
	[1.216, 1.326]	[0.913, 1.021]
Hispanic/Latino	0.956	0.826
	[0.878, 1.040]	[0.731, 0.930]
Ethnicity: Unknown	1.176	1.055
	[1.108, 1.248]	[0.972, 1.144]
Num.Obs.	112 009	87 025
AIC	63 313.9	38 992.1
BIC	63 371.7	39 048.3
Log.Lik.	−31 650.965	−19 490.036

```r
modelplot(
  mods_all,
  exponentiate = TRUE,
  coef_omit = "(Intercept)"
)
```

```
Profiled confidence intervals may take longer time to compute.
  Use `ci_method="wald"` for faster computation of CIs.
Profiled confidence intervals may take longer time to compute.
  Use `ci_method="wald"` for faster computation of CIs.
```



*Possible third analysis, HRF vs compensated hypercapnia - does elevated BMI correlate with progression to respiratory failure in a population that is already hypercapnic

Unadjusted

```r
model_hypercapnia_sub <- glm(
  hrf_vs_comp ~ curr_bmi_num,
  data = hypercapnia_cohort,
  family = binomial
)

summary(model_hypercapnia_sub)
```

```
Call:
glm(formula = hrf_vs_comp ~ curr_bmi_num, family = binomial,
    data = hypercapnia_cohort)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  -1.134760   0.049617  -22.87  < 2e-16 ***
curr_bmi_num  0.010341   0.001618    6.39 1.66e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 37189  on 30280  degrees of freedom
Residual deviance: 37148  on 30279  degrees of freedom
  (95667 observations deleted due to missingness)
AIC: 37152

Number of Fisher Scoring iterations: 4
```

More modest effect but seems to still be statistically significant

Adjusted

```r
model_hypercapnia_sub_adj <- glm(
  hrf_vs_comp ~ curr_bmi_num + age_at_encounter + sex + ethnicity,
```

```r
  data = hypercapnia_cohort,
  family = binomial
)

summary(model_hypercapnia_sub_adj)
```

```
Call:
glm(formula = hrf_vs_comp ~ curr_bmi_num + age_at_encounter +
    sex + ethnicity, family = binomial, data = hypercapnia_cohort)

Coefficients:
                             Estimate Std. Error z value Pr(>|z|)
(Intercept)                -0.8472573  0.0749272 -11.308  < 2e-16 ***
curr_bmi_num                0.0107950  0.0016282   6.630 3.36e-11 ***
age_at_encounter           -0.0062672  0.0007898  -7.935 2.11e-15 ***
sexMale                     0.0968791  0.0256352   3.779 0.000157 ***
ethnicityHispanic or Latino 0.0336114  0.0502445   0.669 0.503523
ethnicityUnknown            0.2130299  0.0355348   5.995 2.04e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 37189  on 30280  degrees of freedom
Residual deviance: 37019  on 30275  degrees of freedom
  (95667 observations deleted due to missingness)
AIC: 37031

Number of Fisher Scoring iterations: 4
```

also holds with adjusted analysis

1.7.4 Are proportions of hypercapnic respiratory failure (v2) different across BMI categories?

```
# Subset to patients with BMI categories
bmi_hrf_strict_subset <- stata_hrf_strict_analysis %>%
  filter(!is.na(bmi_category))

# Summary table by BMI category
bmi_hrf_strict_summary <- bmi_hrf_strict_subset %>%
  group_by(bmi_category) %>%
  summarise(
    n_total = n(),
    n_hrf = sum(hrf_hypercapnic_v2_strict == 1),
    n_no_hrf = sum(hrf_hypercapnic_v2_strict == 0),
    pct_hrf = round(100 * n_hrf / n_total, 1),
    pct_no_hrf = round(100 * n_no_hrf / n_total, 1)
  ) %>%
  arrange(factor(bmi_category, levels = c("Underweight", "Normal weight", "Overweight",
                                          "Obesity I", "Obesity II", "Obesity III")))

# Chi-square test
chisq_test_strict <- chisq.test(table(
  bmi_hrf_strict_subset$bmi_category,
  bmi_hrf_strict_subset$hrf_hypercapnic_v2_strict
))

# View summary table and p-value
bmi_hrf_strict_summary
```

bmi_category	n_total	n_hrf	n_no_hrf	pct_hrf	pct_no_hrf
Underweight	3866	377	3489	9.8	90.2
Normal weight	24496	1301	23195	5.3	94.7
Overweight	25060	1165	23895	4.6	95.4
Obesity I	16721	906	15815	5.4	94.6
Obesity II	9273	666	8607	7.2	92.8

bmi_category	n_total	n_hrf	n_no_hrf	pct_hrf	pct_no_hrf
Obesity III	7609	832	6777	10.9	89.1

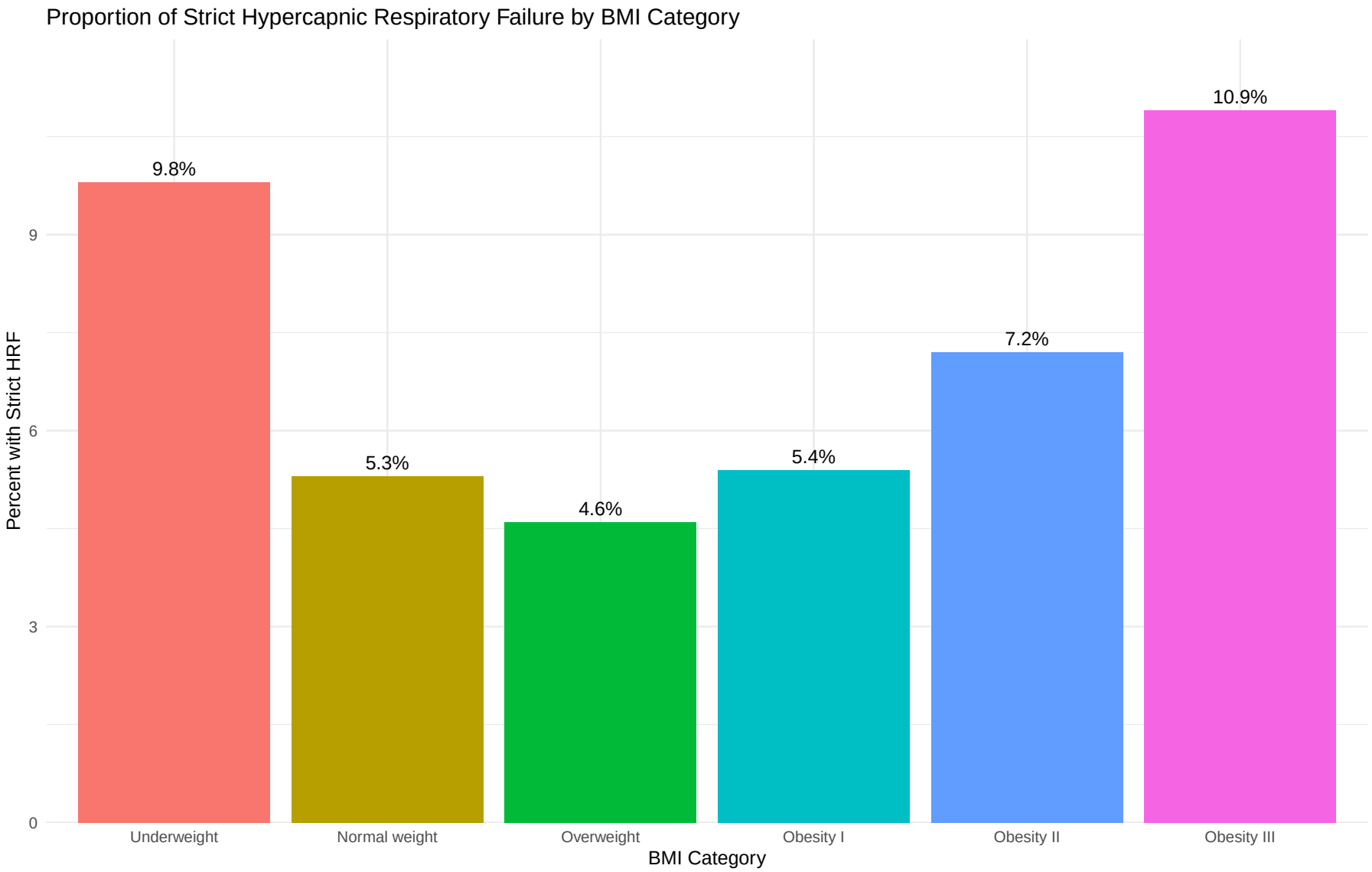
```
chisq_test_strict$p.value
```

```
[1] 3.879776e-118
```

and visualization

```
# Make sure bmi_category is ordered
bmi_hrf_strict_summary <- bmi_hrf_strict_summary %>%
  mutate(bmi_category = factor(bmi_category,
                               levels = c("Underweight", "Normal weight", "Overweight",
                                          "Obesity I", "Obesity II", "Obesity III")))

# Plot proportion of strict HRF by BMI category
ggplot(bmi_hrf_strict_summary, aes(x = bmi_category, y = pct_hrf, fill = bmi_category)) +
  geom_col() +
  geom_text(aes(label = paste0(pct_hrf, "%")), vjust = -0.5) +
  scale_y_continuous(expand = expansion(mult = c(0, 0.1))) +
  labs(
    title = "Proportion of Strict Hypercapnic Respiratory Failure by BMI Category",
    x = "BMI Category",
    y = "Percent with Strict HRF"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```



1.8 Section 7. Population attributable fraction

*this is my first pass at the PAF with the adjusted primary cohort

Everyone set to BMI 25

```
stata_hrf_analysis$pred_obs <- predict(baseline_model, type = "response")

risk_observed <- mean(stata_hrf_analysis$pred_obs, na.rm = TRUE)
risk_observed
```

```
[1] 0.08213626
```

```
stata_cf <- stata_hrf_analysis
stata_cf$curr_bmi_num <- 25
```

```
stata_cf$pred_cf <- predict(baseline_model, newdata = stata_cf, type = "response")

risk_cf_25 <- mean(stata_cf$pred_cf, na.rm = TRUE)
risk_cf_25
```

```
[1] 0.07598445
```

```
paf_25 <- (risk_observed - risk_cf_25) / risk_observed
paf_25
```

```
[1] 0.07489761
```

Everyone set to a BMI of 30

```
stata_hrf_analysis$pred_obs <- predict(baseline_model, type = "response")

risk_observed <- mean(stata_hrf_analysis$pred_obs, na.rm = TRUE)
risk_observed
```

```
[1] 0.08213626
```

```r
stata_cf_30 <- stata_hrf_analysis
stata_cf_30$curr_bmi_num <- 30
```

```r
stata_cf_30$pred_cf_30 <- predict(
  baseline_model,
  newdata = stata_cf_30,
  type = "response"
)

risk_cf_30 <- mean(stata_cf_30$pred_cf_30, na.rm = TRUE)
risk_cf_30
```

```
[1] 0.08310813
```

```r
paf_30 <- (risk_observed - risk_cf_30) / risk_observed
paf_30
```

```
[1] -0.01183242
```

Now with everyone over 30 reduced to 29.9 (keeping those below 29.9 they same BMI) - this is probably what most accurately captures our study question

```r
stata_hrf_analysis$pred_obs <- predict(
  baseline_model,
  type = "response"
)

risk_observed <- mean(
  stata_hrf_analysis$pred_obs,
  na.rm = TRUE
)

risk_observed
```

```
[1] 0.08213626
```

```
stata_cf_no_obesity <- stata_hrf_analysis

stata_cf_no_obesity$curr_bmi_num <- ifelse(
  stata_cf_no_obesity$curr_bmi_num >= 30,
  29.9,
  stata_cf_no_obesity$curr_bmi_num
)

stata_cf_no_obesity$pred_cf_no_obesity <- predict(
  baseline_model,
  newdata = stata_cf_no_obesity,
  type = "response"
)

risk_cf_no_obesity <- mean(
  stata_cf_no_obesity$pred_cf_no_obesity,
  na.rm = TRUE
)

risk_cf_no_obesity
```

```
[1] 0.07816554
```

```
stata_cf$pred_cf <- predict(
  baseline_model,
  newdata = stata_cf,
  type = "response"
)

risk_counterfactual <- mean(
  stata_cf$pred_cf,
  na.rm = TRUE
)
```

```
risk_counterfactual
```

```
[1] 0.07598445
```

```
paf_no_obesity <- (
  risk_observed - risk_cf_no_obesity
) / risk_observed

paf_no_obesity
```

```
[1] 0.04834304
```

if everyone were set 18.5 to 25 (probably this doesn't accurately reflect our study question, because it also shifts underweight patients up, but interesting):

```
stata_cf_normal <- stata_hrf_analysis

stata_cf_normal$curr_bmi_num <- ifelse(
  stata_cf_normal$curr_bmi_num > 25,
  25,
  ifelse(
    stata_cf_normal$curr_bmi_num < 18.5,
    18.5,
    stata_cf_normal$curr_bmi_num
  )
)
```

```
stata_cf_normal$pred_cf_normal <- predict(
  baseline_model,
  newdata = stata_cf_normal,
  type = "response"
)
```

```r
risk_cf_normal <- mean(
  stata_cf_normal$pred_cf_normal,
  na.rm = TRUE
)

risk_cf_normal
```

```
[1] 0.07458487
```

```r
paf_normal <- (
  risk_observed - risk_cf_normal
) / risk_observed

paf_normal
```

```
[1] 0.0919373
```

PAF table

```r
paf_results <- data.frame(
  scenario = c(
    "Everyone set to BMI 30",
    "Everyone set to BMI 25",
    "Obesity eliminated (BMI >=30 -> 29.9)",
    "All BMI values shifted into normal range (18.5-25)"
  ),

  observed_risk = c(
    risk_observed,
    risk_observed,
    risk_observed,
    risk_observed
  ),
```

```
  counterfactual_risk = c(
    risk_cf_30,
    risk_cf_25,
    risk_cf_no_obesity,
    risk_cf_normal
  ),

  paf_percent = c(
    paf_30,
    paf_25,
    paf_no_obesity,
    paf_normal
  ) * 100
)
```

```
paf_results$paf_percent <- round(paf_results$paf_percent, 2)
paf_results$observed_risk <- round(paf_results$observed_risk, 4)
paf_results$counterfactual_risk <- round(paf_results$counterfactual_risk, 4)

paf_results
```

scenario	observed_risk	counterfactual_risk	paf_percent
Everyone set to BMI 30	0.0821	0.0831	-1.18
Everyone set to BMI 25	0.0821	0.0760	7.49
Obesity eliminated (BMI >=30 -> 29.9)	0.0821	0.0782	4.83
All BMI values shifted into normal range (18.5-25)	0.0821	0.0746	9.19

Confidence intervals:

Below is the way I approached creating a confidence interval with bootstrapping. Seems there's a few ways to do this, ended up using a fixed model. I used the PAF model above for "eliminated obesity" (everyone >30 set to 29.9)

```
set.seed(123)
```

```r
n_boot <- 1000

paf_boot <- numeric(n_boot)
```

```r
for (b in 1:n_boot) {

  # resample rows
  boot_idx <- sample(
    1:nrow(stata_hrf_analysis),
    replace = TRUE
  )

  boot_data <- stata_hrf_analysis[boot_idx, ]

  # observed risk
  boot_obs_pred <- predict(
    baseline_model,
    newdata = boot_data,
    type = "response"
  )

  risk_obs <- mean(boot_obs_pred, na.rm = TRUE)

  # counterfactual: eliminate obesity
  boot_cf <- boot_data

  boot_cf$curr_bmi_num <- ifelse(
    boot_cf$curr_bmi_num >= 30,
    29.9,
    boot_cf$curr_bmi_num
  )

  boot_cf_pred <- predict(
    baseline_model,
    newdata = boot_cf,
    type = "response"
```

```
  )

  risk_cf <- mean(boot_cf_pred, na.rm = TRUE)

  # PAF
  paf_boot[b] <- (risk_obs - risk_cf) / risk_obs
}
```

```
paf_mean <- mean(paf_boot, na.rm = TRUE)

paf_ci <- quantile(
  paf_boot,
  probs = c(0.025, 0.975),
  na.rm = TRUE
)

paf_mean
```

```
[1] 0.0483345
```

```
paf_ci
```

```
      2.5%      97.5%
0.04784395 0.04880581
```

```
paf_results_boot <- data.frame(
  paf_percent = paf_boot * 100
)

mean(paf_results_boot$paf_percent, na.rm = TRUE)
```

```
[1] 4.83345
```

```r
quantile(paf_results_boot$paf_percent, c(0.025, 0.975), na.rm = TRUE)
```

```
    2.5%    97.5%
4.784395 4.880581
```

1.9 Appendix

```r
sessionInfo()
```

```
R version 4.5.2 (2025-10-31 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 26200)

Matrix products: default
  LAPACK version 3.12.1

locale:
[1] LC_COLLATE=English_United States.utf8
[2] LC_CTYPE=English_United States.utf8
[3] LC_MONETARY=English_United States.utf8
[4] LC_NUMERIC=C
[5] LC_TIME=English_United States.utf8

time zone: America/Denver
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
[1] tidyr_1.3.1        dplyr_1.1.4        stringr_1.6.0      gtsummary_2.5.0
[5] here_1.0.2         modelsummary_2.6.0 ggplot2_4.0.1      haven_2.5.5
```

```
loaded via a namespace (and not attached):
 [1] gt_1.2.0           generics_0.1.4     xml2_1.5.1         stringi_1.8.7
 [5] hms_1.1.4          digest_0.6.38      magrittr_2.0.4     evaluate_1.0.5
 [9] grid_4.5.2         RColorBrewer_1.1-3 cards_0.7.1        fastmap_1.2.0
[13] rprojroot_2.1.1    jsonlite_2.0.0     backports_1.5.0    purrr_1.2.0
[17] scales_1.4.0       cli_3.6.5          rlang_1.1.6        performance_0.16.0
[21] litedown_0.8       commonmark_2.0.0   tinytable_0.16.0   withr_3.0.2
[25] yaml_2.3.10        datawizard_1.3.0   tools_4.5.2        checkmate_2.3.4
[29] bayestestR_0.17.0  forcats_1.0.1      vctrs_0.6.5        R6_2.6.1
[33] lifecycle_1.0.4    fs_1.6.6           insight_1.4.6      pkgconfig_2.0.3
[37] pillar_1.11.1      gtable_0.3.6       data.table_1.17.8  glue_1.8.0
[41] xfun_0.54          tibble_3.3.0       tidyselect_1.2.1   parameters_0.28.3
[45] rstudioapi_0.17.1  knitr_1.50         farver_2.1.2       htmltools_0.5.8.1
[49] tables_0.9.33      rmarkdown_2.30     labeling_0.4.3     compiler_4.5.2
[53] S7_0.2.1           markdown_2.0
```